Kenbunshoku

Project Plan

Global AI Hackathon Series with Qwen Cloud — Track 5: EdgeAgent

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1. Problem Statement

Existing doorbell/security cameras record and alert on raw motion, forcing homeowners to review footage after the fact or get flooded with low-value notifications. They don't distinguish a familiar, expected visitor from an unfamiliar or anomalous one, and they carry no memory of prior visits.

2. Goal

Build a camera-agnostic agent that ingests a live video stream from any connected camera, detects approaching people, reasons about context using Qwen Cloud (Qwen-VL), remembers visitor patterns over time, and sends the homeowner a plain-language, context-aware alert — never an autonomous action.

3. Scope

3.1 In scope (MVP)

- Camera-agnostic stream ingestion (tested via phone acting as a simulated IP camera)
- Local/edge motion and person detection (YOLO), packaged as a portable Docker service
- Cloud reasoning via Qwen-VL, hosted on Alibaba Cloud, for visitor context classification
- Lightweight memory/pattern store for recurring-visitor recognition
- Push notification to homeowner with the agent's reasoning
- Graceful fallback to a generic "person detected" alert if cloud is unreachable
- All required Devpost submission artifacts: public licensed repo, architecture diagram, proof of Alibaba Cloud deployment, 3-minute demo video, project write-up

3.2 Explicitly out of scope

- Weapon/threat detection
- Any autonomous action (locking doors, contacting authorities, sirens)
- LiDAR-dependent 3D rendering
- Multi-camera/dashcam fusion
- Face-recognition-level identity matching

4. Target Track & Judging Fit

Primary track: EdgeAgent (Track 5) — edge sensing via the camera/edge service, cloud reasoning via Qwen Cloud, and human-in-the-loop action. Secondary theme: pattern memory (echoing Track 1's MemoryAgent spirit) is used as a supporting differentiator, not a second full submission.

5. Timeline (Jul 15 – Jul 20, 2026 · deadline 2:00pm Pacific)

Date	Focus	Deliverable checkpoint
Jul 15	Finalize scope, register Qwen Cloud, claim voucher, set up repo skeleton	Repo created, Qwen Cloud account active
Jul 16	Build edge service: motion/person detection (YOLO) + phone-as-camera test stream	Local detection working on a live phone feed
Jul 17	Stand up Alibaba Cloud backend: ingestion API + Qwen-VL call + basic memory store	End-to-end: phone → edge → cloud → classification returned
Jul 18	Build alert/notification layer + memory pattern logic (recurring visitor detection)	Full loop working: detection → reasoning → alert on phone
Jul 19	Polish, record proof-of-Alibaba-Cloud-deployment clip, build architecture diagram, write submission text	All required submission artifacts drafted
Jul 20	Final testing, record 3-min demo video, submit before 2:00pm Pacific	Submitted on Devpost

6. Roles

Solo build by Daniel Ainoko (Flutter/mobile engineer). If a teammate joins, natural split is: (a) edge service + detection, (b) Alibaba Cloud backend + Qwen integration, (c) notification client + submission materials.

7. Risks & Mitigations

Risk	Mitigation
Weapon/threat detection scope creep eats the timeline	Explicitly parked as future work; MVP stays on context classification
Alibaba Cloud deployment proof missed at the last minute	Build and record proof by Jul 19, not on submission day
Phone-as-camera stream flaky (RTSP/IP-cam app issues)	Test the camera-simulation path on Day 1 before building anything downstream
Qwen Cloud voucher/coupon delays	Claim voucher immediately; coupon cutoff has already passed for new requests so use existing free-trial credits
Over-scoping across multiple tracks	Commit to EdgeAgent as primary track in writing before building

8. Deliverables Checklist

- Public GitHub repo with visible open-source license

- Architecture diagram
- Proof-of-Alibaba-Cloud-deployment recording + linked code file
- 3-minute demo video (YouTube/Vimeo, public)
- Project text description identifying the track
- (Optional) Public blog/social post documenting the build, for the Blog Post bonus prize